

SOLAR ASTROPHYSICS SCHOOL · OCTOBER 2026

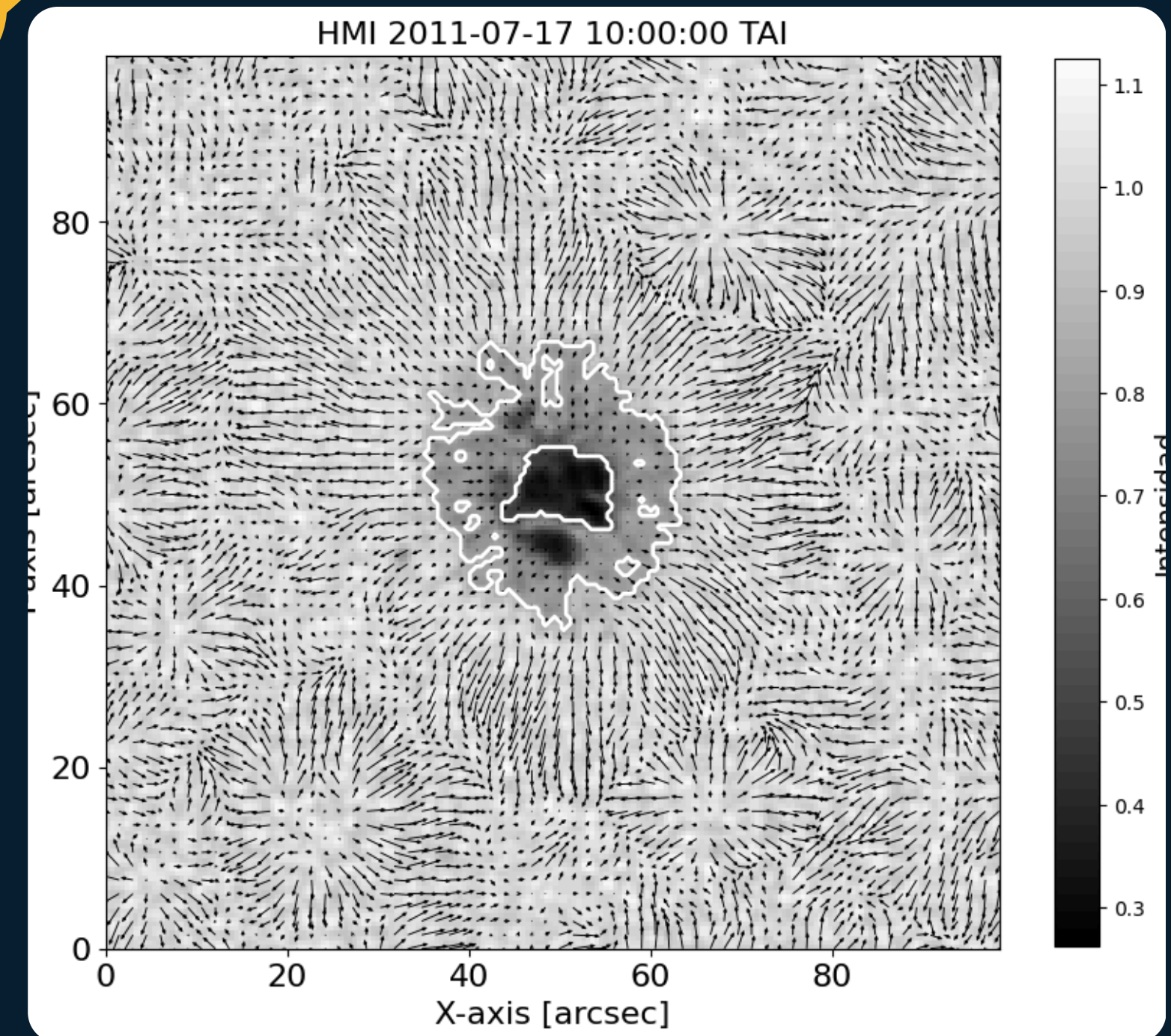
HelioData Academy 2026



Sunspots

Working with solar data in sunspots context

Juan Pablo Herrera · Observatorio Astronómico Nacional · October 8 2025



Group of Solar Astrophysics GoSA
Observatorio Astronómico Nacional · UNAL Bogotá

Agenda

SESSION ROUTE



01

Sunspots



Umbra and penumbra, magnetic structure

02

Continuum and LOS magnetic field data and images



FITS archives in sunspot context: reading the header and plot the data.

03

Working with sunspots



Area of the umbras and penumbras, magnetic profiles of sunspots,
Evolution of a sunspot size.

Agenda

RUTA DE LA SESIÓN



04

Tracking flows around sunspots: LCT



How LCT works, the parameters of a datacube:
cadence, pixel scale and apodization window

05

The moat flows



The direction of flows around sunspots

06

Exercises



Apply LCT in a sunspot datacube

SECTION

01

Sunspots

Structure and magnetic profile



CONCEPTO

Sunspots

Sunspots are dark patches in the solar photosphere, caused by a strong concentration of magnetic field

- The strong magnetic field reduces the temperature in the photospheric region, causing this region to appear darker.
- Sunspots are characterized by a dark central umbra surrounded by a brighter penumbra.

IDEA CLAVE

Sunspots are dark areas caused by magnetic field

Sunspot

Umbra

Penumbra

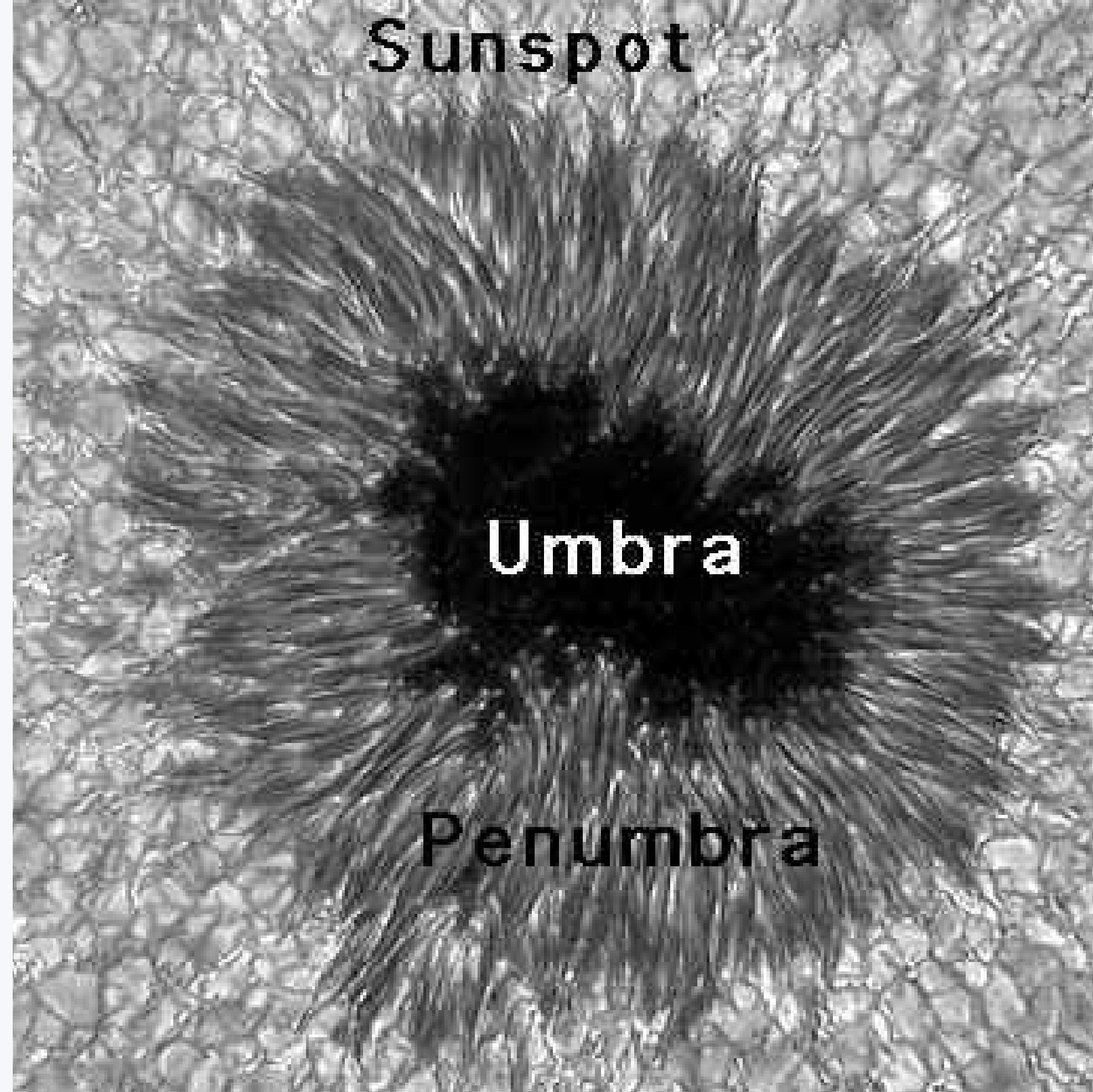
CONCEPTO

Sunspot temperatures

- Umbra: ~ 3500 K
- Penumbra: ~ 4000 K
- Quiet Sun: ~ 5700 K

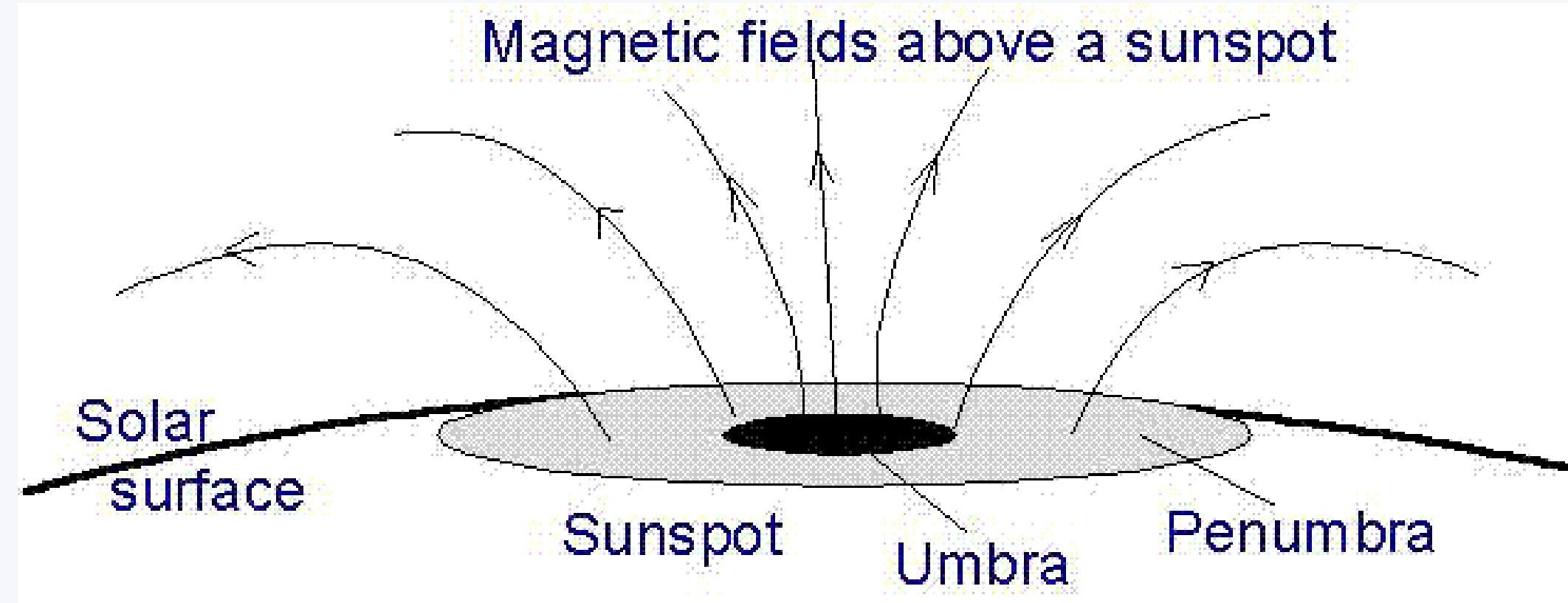
IDEA CLAVE

The temperatures in the sunspots are lower than quiet Sun



Magnetic structure

The magnetic field lines at the center of the umbra are predominantly vertical and exhibit the highest field strength. Moving outward from this region, the field lines become increasingly inclined and weaker.



IDEA CLAVE

Sunspots exhibit a well-defined magnetic structure.

SECTION

02

Continuum and LOS magnetic field data with SDO/HMI

FITS archives in sunspot context: reading the header and plot the data.



2 types of data



Continuum vs LOS magnetograms

Continuum

Shows the structures in the photosphere
We can track flows and evolution of granules and sunspots

Magnetograms LOS

Shows the LOS component of magnetic vector
We can see the places with strong or weak magnetic activity, with positive or negative polarity

With both filtergrams we can study the sunspots dynamics

CONCEPTO

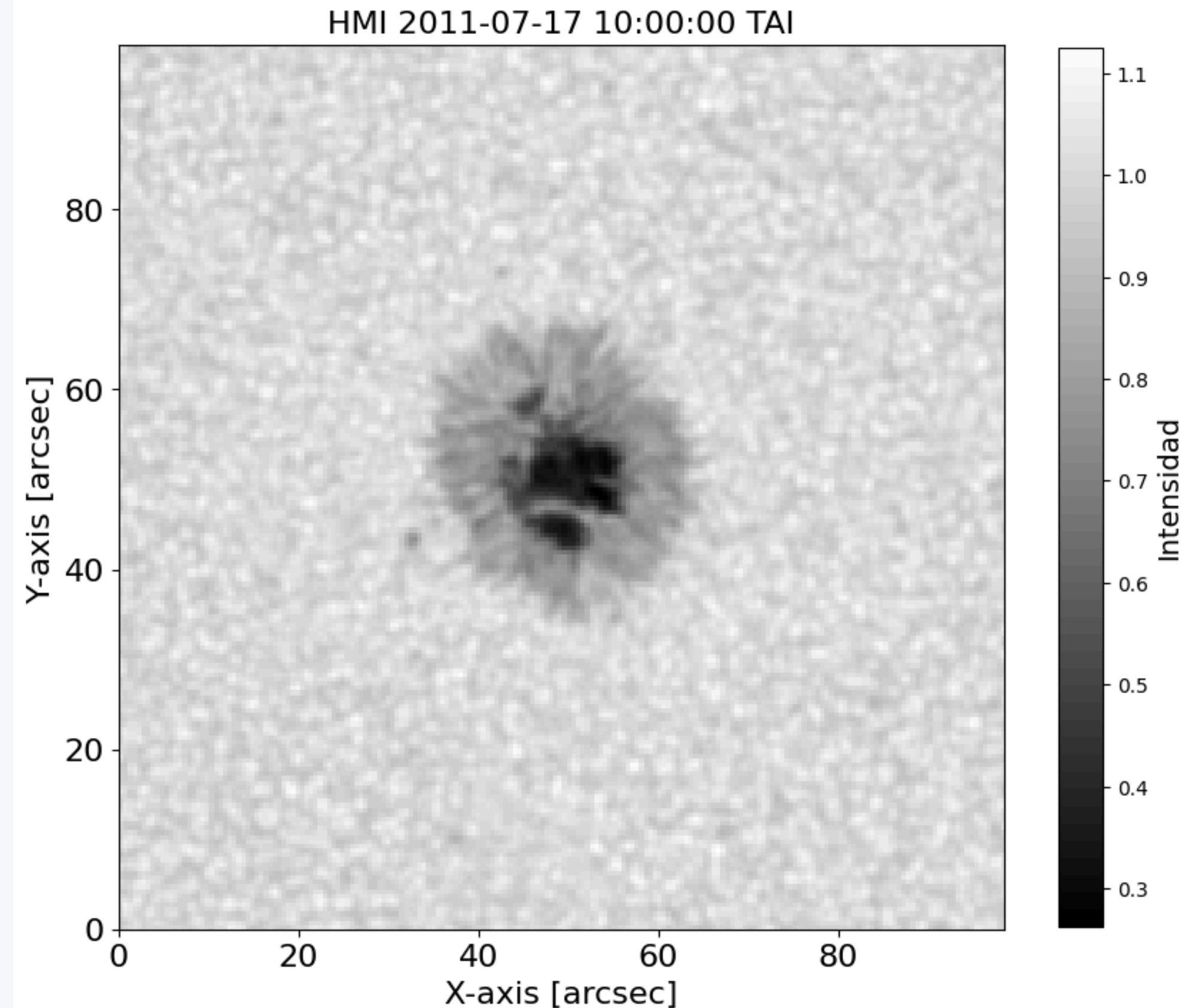
Field of View (FOV) frame

The origin of the helioprojective coordinate system can be redefined from the center of the solar disk to the region of interest under investigation within the image.

- We use the The NAXIS and CDELT header cards

IDEA CLAVE

We will build the FOV frame



CONCEPTO

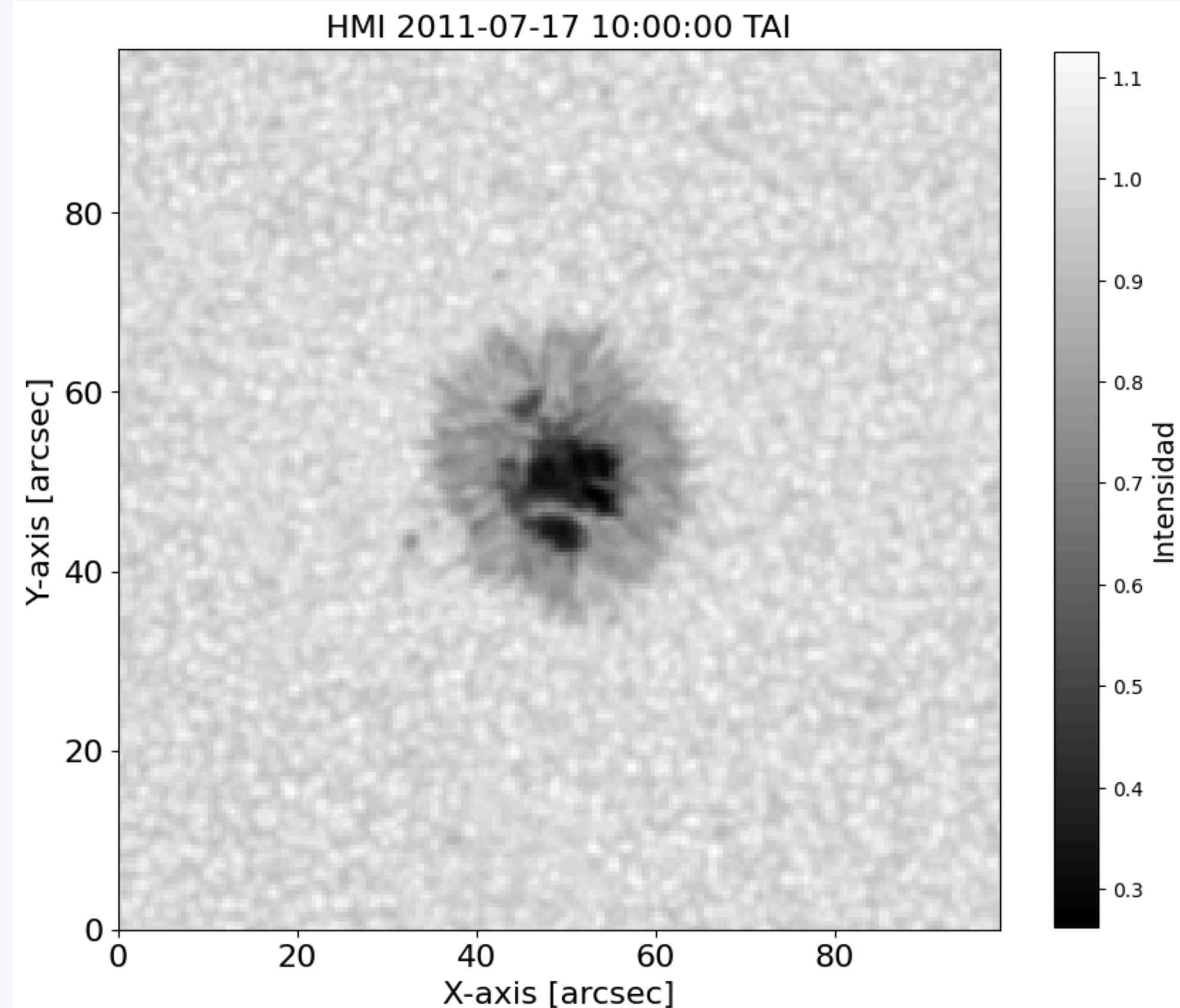
The normalized intensity

Rather than using the absolute intensity, we employ the normalized continuum intensity, defined relative to the mean intensity of the quiet Sun.

We can easily mask sunspots with this method.

IDEA CLAVE

In continuum data we should normalize the intensity values.

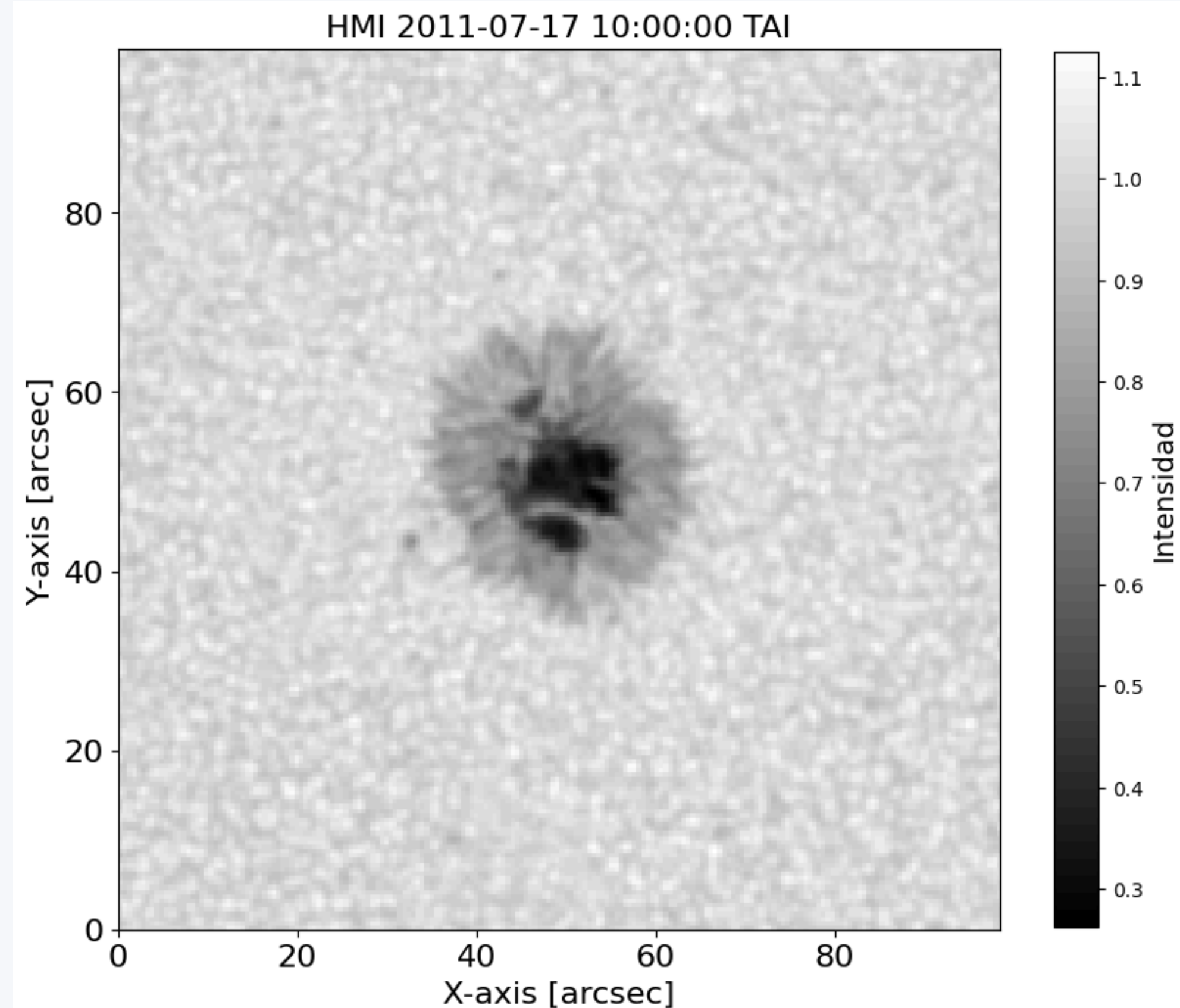


NORMALIZED INTENSITY

$$I_N = I / I_0$$

Normalized Intensity

The normalized intensity is obtained by dividing the absolute intensity by the mean intensity of the quiet Sun.



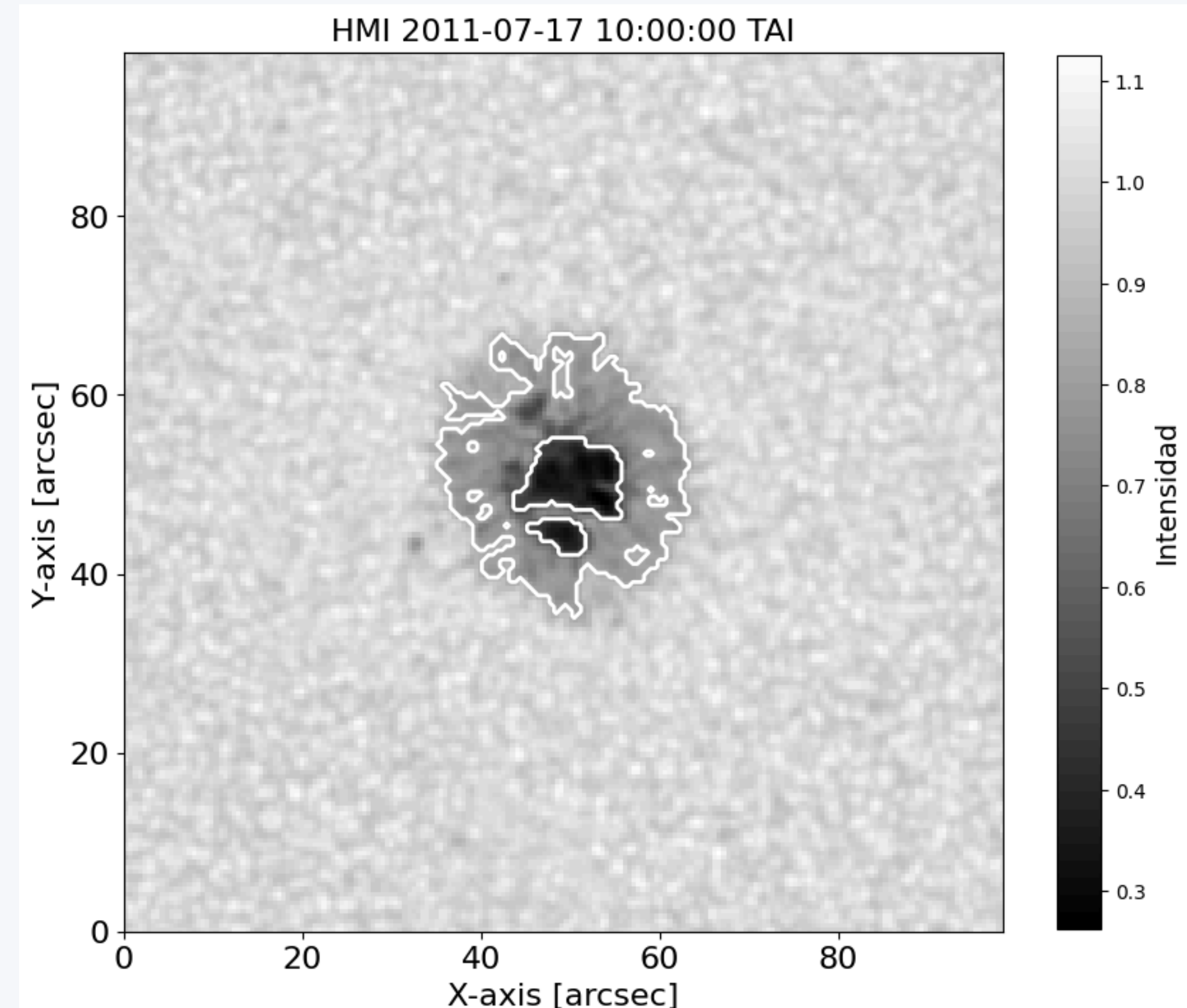
NORMALIZED INTENSITY

$$I_u = 0.55 I_N$$

$$I_p = 0.85 I_N$$

Normalized Intesity

Suitable intensity thresholds for distinguishing the umbra and penumbra are approximately ($0.55 I_N$) and ($0.85 I_N$), respectively.



CONCEPTO

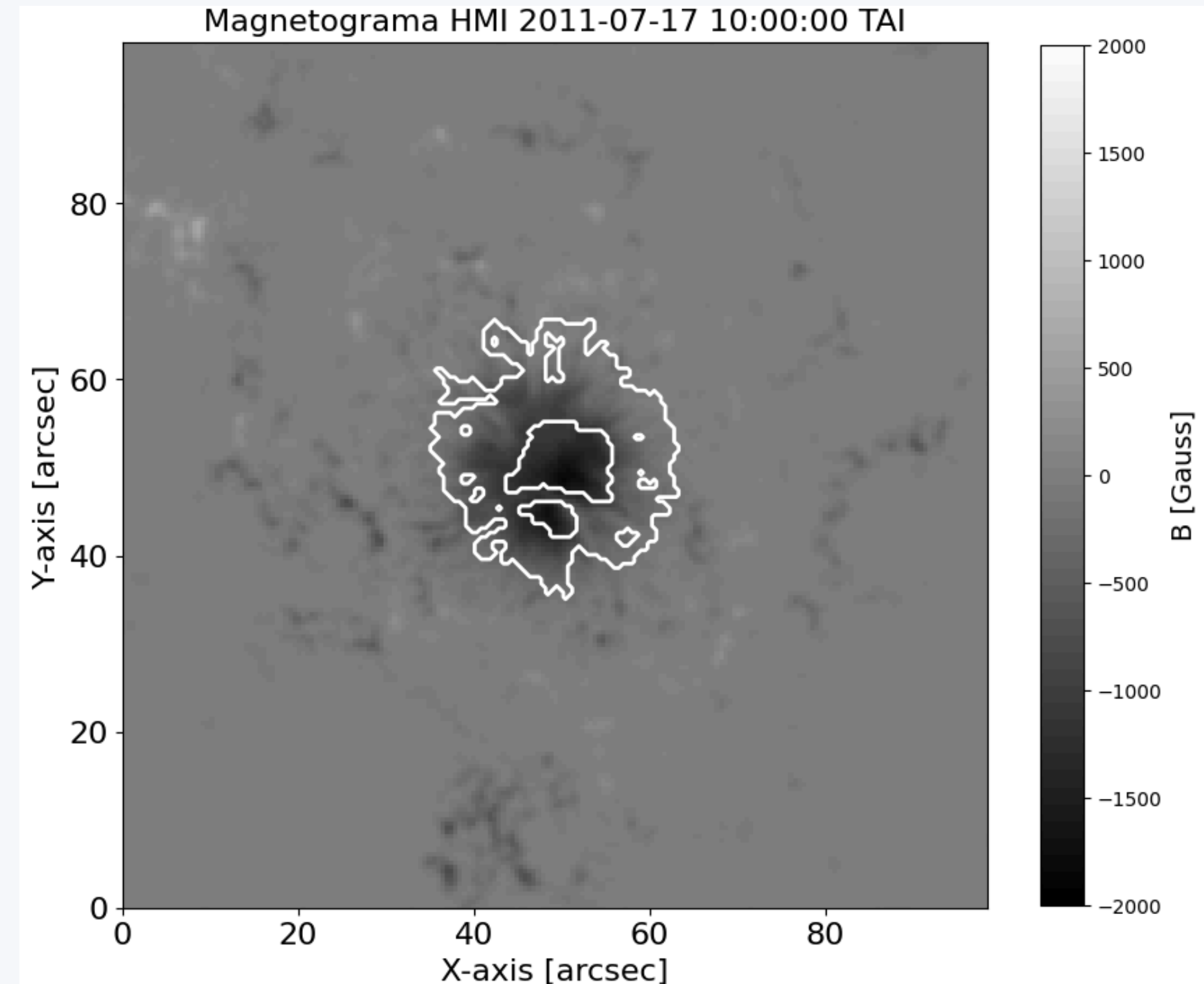
LOS magnetogram

A clear concentration of the magnetic field in the interior of the sunspot can be observed.

This is only a component of full vector magnetic field, for obtain them, we need more complex data.

IDEA CLAVE

LOS Magnetograms shows the basic magnetic structure of a sunspot



SECTION

03

Working with sunspots

Area of the umbras and penumbras, magnetic profiles of sunspots,
Evolution of a sunspot size.



Hands-on 1: Area and magnetic profile

1. Plot graphs with FOV frame
2. Mask the umbra and penumbra
3. Measure the area of sunspot

OBJETIVO

Measure the area of a umbra, penumbra and a full sunspot and make a B profile.

```
[nombre_del_archivo].py

1 for thresh in [0.55, 0.85]:
2     mask = I2 / I2_QS < thresh * np.mean(I2/I2_QS)
3     labeled_mask = label(mask)
4     regions = regionprops(labeled_mask)
5     if regions:
6         largest_region = max(regions, key=lambda r: r.area)
7         pixel_area = 0.50433 ** 2 # arcsec2 por píxel
8         area_arcsec2 = largest_region.area * pixel_area
9         print(area_arcsec2)
10    mask_sunspot = labeled_mask == largest_region.label
11    ax.contour(x_phys, y_phys, mask_sunspot.astype(float),
               levels=[0.5], colors='w', linewidths=2)
```

EXIT> Spot image with the umbra and penumbra mask

Hands-on 1: Area and magnetic profile

1. Plot the magnetogram with the sunspot mask
2. Make the magnetic profile

OBJETIVO

Measure the area of a umbra, penumbra and a full sunspot and make a B profile.

[nombre_del_archivo].py

```
1 for i in range(len(r_centers)):
2     ring = (
3         (radius >= r_edges[i]) &
4         (radius < r_edges[i + 1])
5     )
6     values = B_LOS[ring]
7     values = values[np.isfinite(values)]
8     if values.size > 0:
9         B_radial_mean[i] = np.mean(values)
10        B_radial_std[i] = np.std(values)
11        N_points[i] = values.size
12    valid = np.isfinite(B_radial_mean)
```

EXIT> Magnetic profile

SECTION

04

Tracking flows around sunspots: LCT



How LCT works, the parameters of a datacube: cadence, pixel scale and apodization window

The datacube

Some FITS files contain multidimensional arrays composed of multiple images. These datasets are known as data cubes, where the third dimension can represent quantities such as time, wavelength, or polarization.

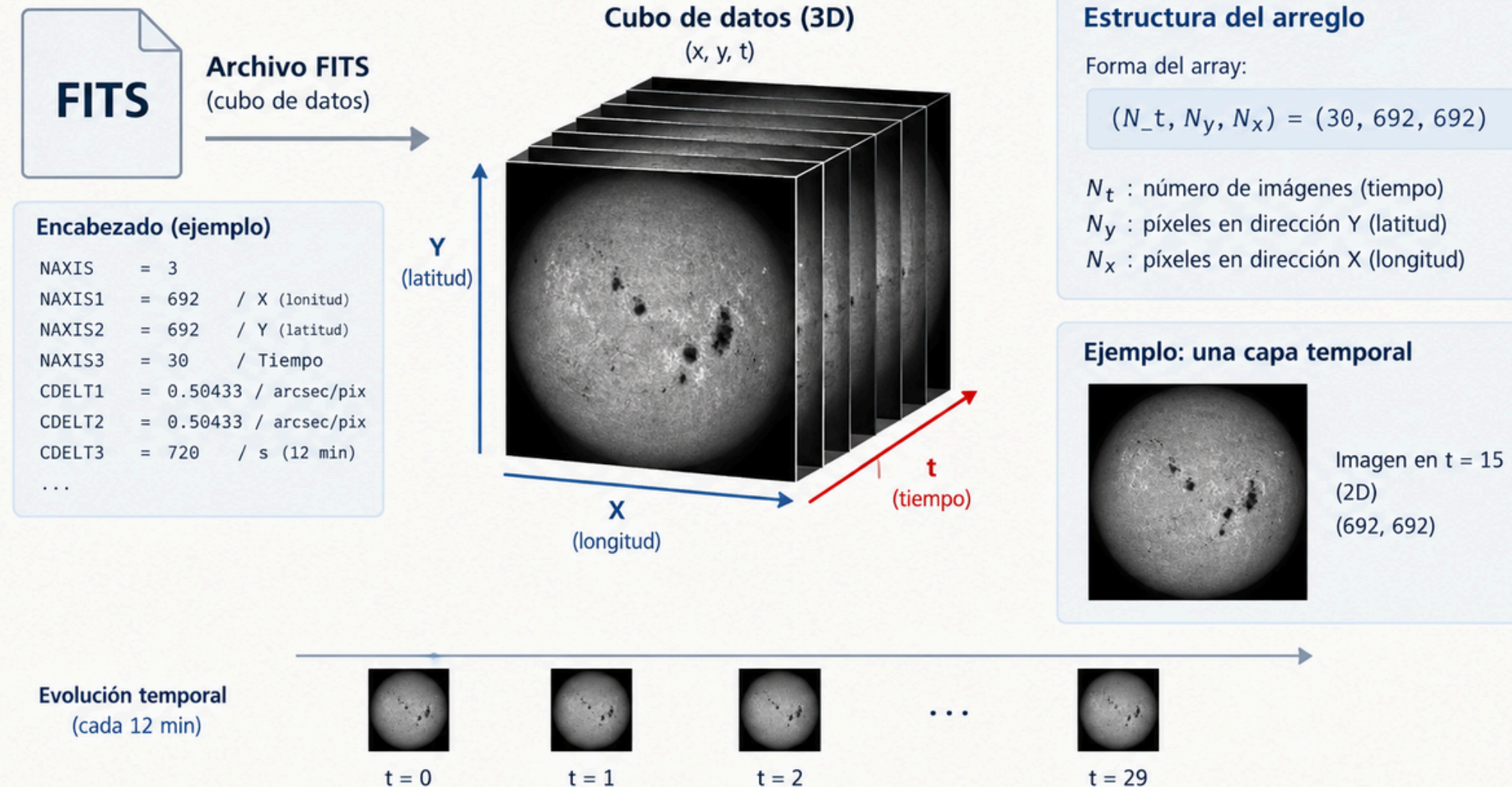
In our case, the quantity is **time**

IDEA CLAVE

A datacube contains more than a single image

FITS 3D Temporal

Un archivo FITS puede contener un cubo de datos, es decir, un conjunto de imágenes organizadas en tres dimensiones. En un cubo temporal, la tercera dimensión corresponde al tiempo.



The datacube: Fundamental cards

The cards we will use next:

- NAXIS1, NAXIS2: Image dimensions
- CDELT1, CDELT2: Arcsecs per pixel
- CADENCE: Time between images

IDEA CLAVE

A datacube contains more than a single image

FITS 3D Temporal

Un archivo FITS puede contener un cubo de datos, es decir, un conjunto de imágenes organizadas en tres dimensiones. En un cubo temporal, la tercera dimensión corresponde al tiempo.



FITS

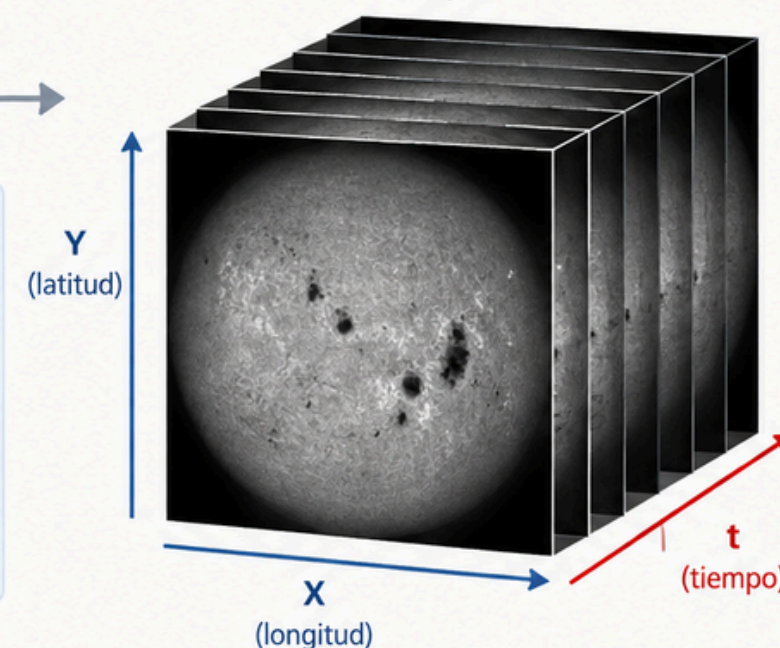
Archivo FITS
(cubo de datos)

Encabezado (ejemplo)

```
NAXIS      = 3
NAXIS1     = 692 / X (lonitud)
NAXIS2     = 692 / Y (latitud)
NAXIS3     = 30 / Tiempo
CDELT1     = 0.50433 / arcsec/pix
CDELT2     = 0.50433 / arcsec/pix
CDELT3     = 720 / s (12 min)
...
```

Cubo de datos (3D)

(x, y, t)



Estructura del arreglo

Forma del array:

$(N_t, N_y, N_x) = (30, 692, 692)$

N_t : número de imágenes (tiempo)

N_y : píxeles en dirección Y (latitud)

N_x : píxeles en dirección X (longitud)

Ejemplo: una capa temporal

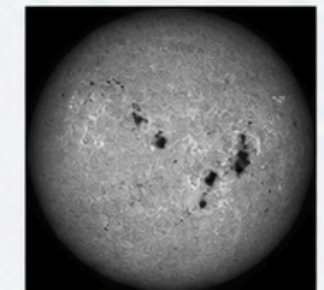
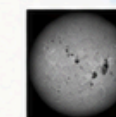
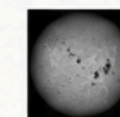


Imagen en $t = 15$
(2D)
(692, 692)

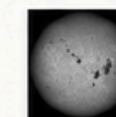
Evolución temporal
(cada 12 min)



t = 0



t = 1



t = 2

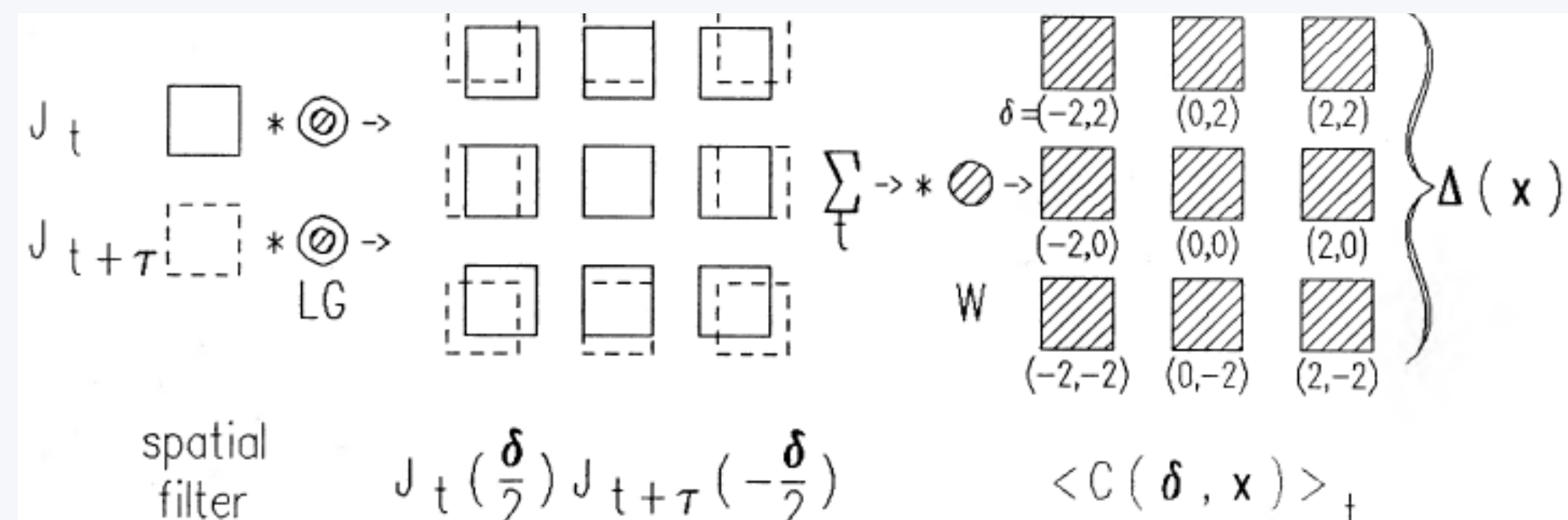
...



t = 29

Local Correlation Tracking (LCT)

Local Correlation Tracking (LCT) is an image-based technique used to estimate horizontal plasma motions by tracking the displacement of localized intensity features between consecutive images.



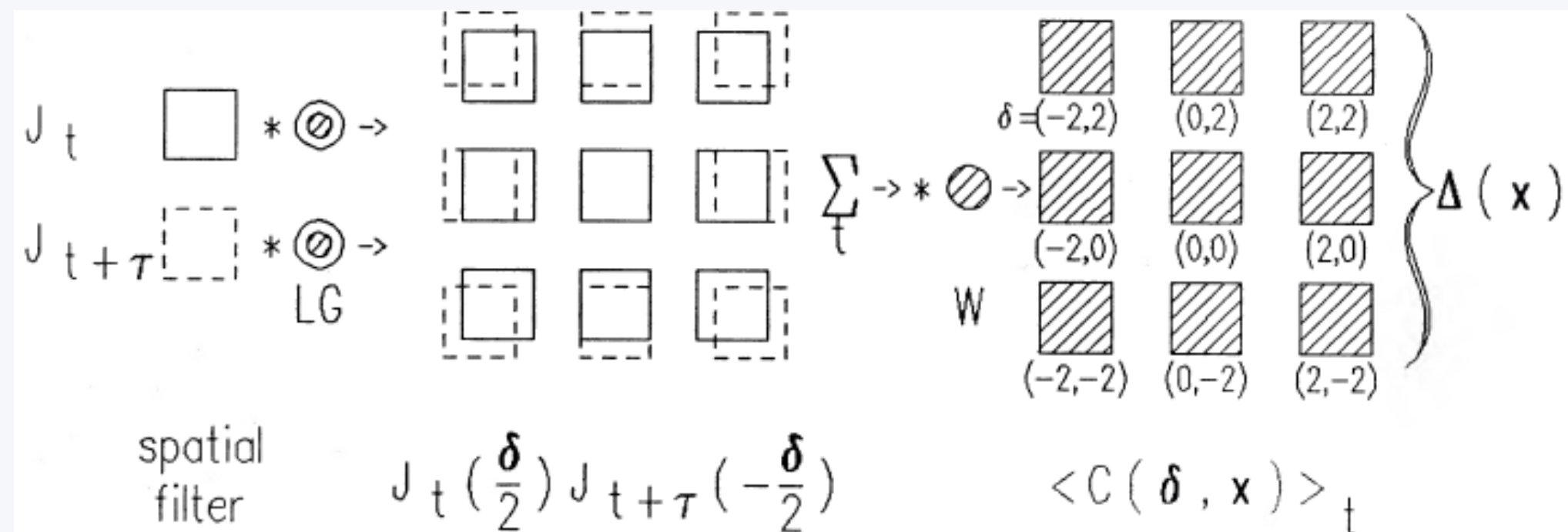
IDEA CLAVE

We use LCT to obtain flows for a continuum datacube

$$C(\delta, x) = \int J_t \left(\xi - \frac{\delta}{2} \right) J_{t+\tau} \left(\xi + \frac{\delta}{2} \right) W(x - \xi) d\xi$$

LOCAL CORRELATION

- J= the consecutive images
- W= An apodization window
- ξ = Space
- δ = displacement of motion
- x= position of W



November & Simon. Astropysical Journal 1988

SECTION

05

The moat flows

The direction of flows around sunspots



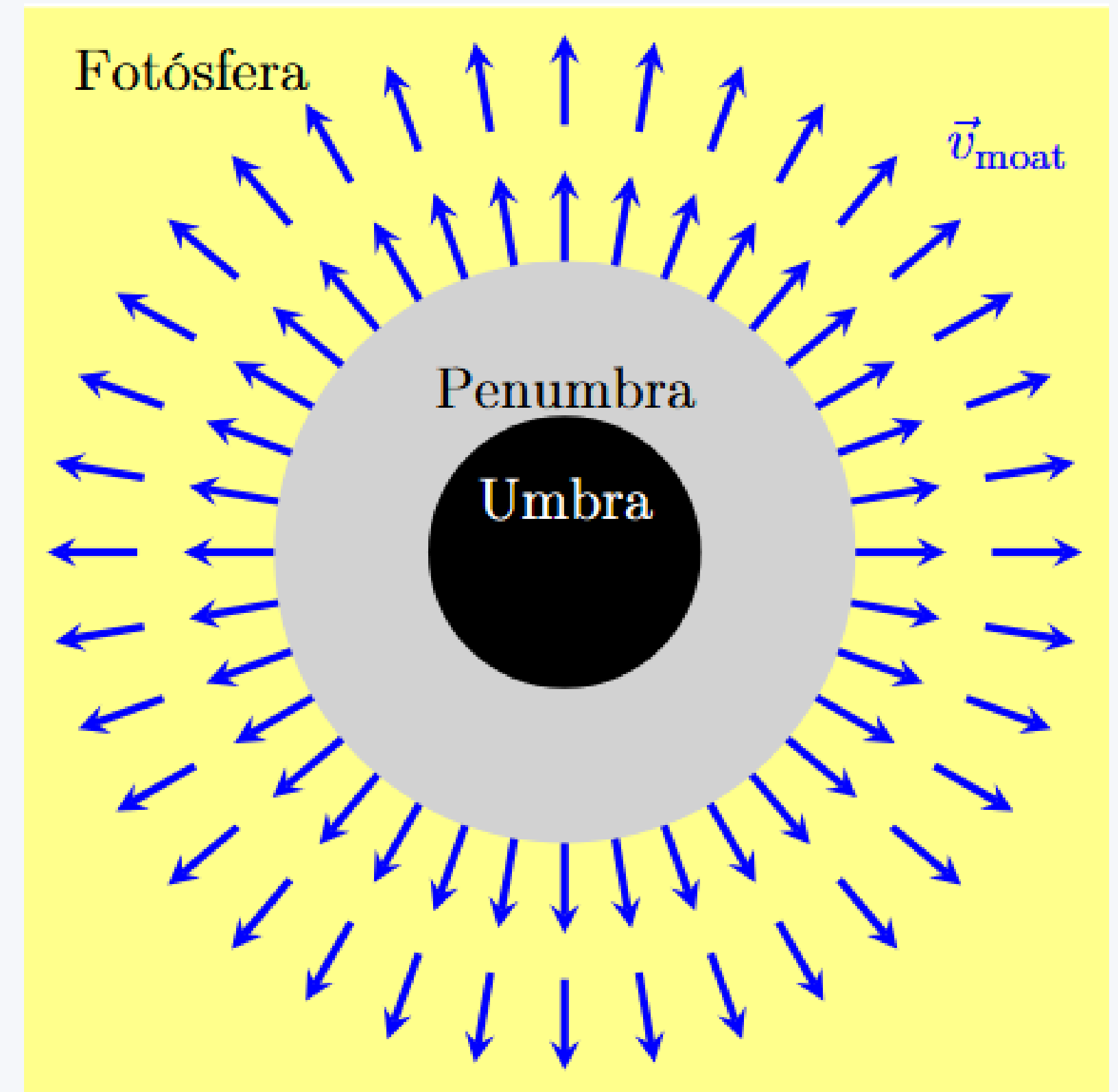
Moat flows

The LCT results shows a pattern of radial flows around the sunspots, this is called the moat flows.

Moat flows are outward-directed horizontal plasma motions surrounding sunspots, extending from the penumbra into the surrounding photosphere.

IDEA CLAVE

We use LCT to obtain flows for a continuum datacube

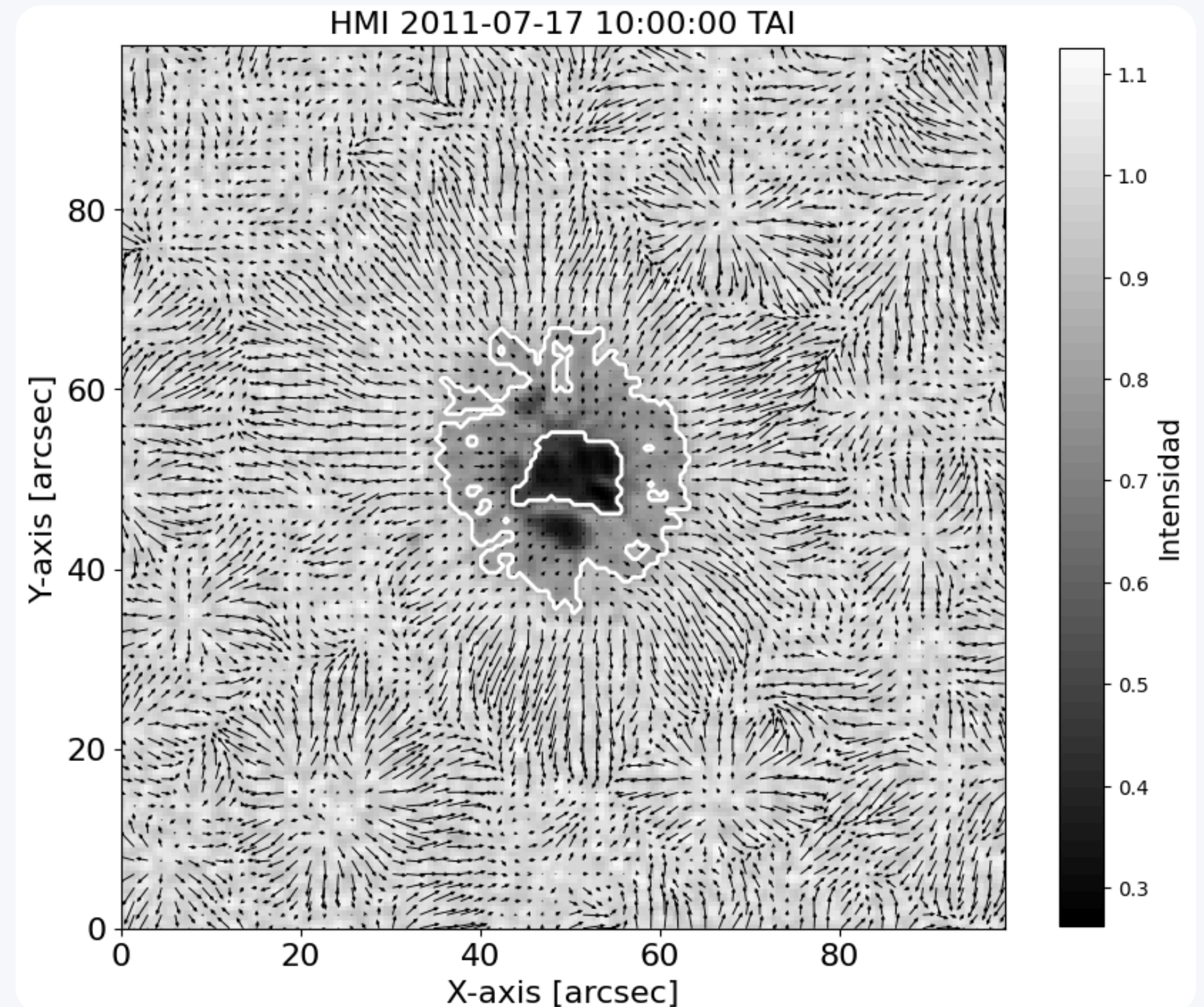


Moat flows

The spatial extent and magnitude of moat flows remain under debate, with studies reporting different flow speeds and radial extents. A general consensus on their properties has yet to be established.

IDEA CLAVE

We use LCT to obtain flows for a continuum datacube



SECTION

06

Exercises

Area of the umbras and penumbras, magnetic profiles of sunspots,
Evolution of a sunspot size.



Hands-on 2: LCT and moat flows

1. Build a datacube
2. Evolution of sunspot size
3. Apply LCT
4. Detect the moat flow

OBJETIVO

Measure flows with LCT, identify the moat flow

```
[nombre_del_archivo].py

1 import numpy as np
2 import matplotlib.pyplot as plt
3 flows2 = flowLCT(data_cube2[0].data, 3, 0.50433, 45)
4 vv2 = np.sqrt(flows2.vx**2 + flows2.vy**2)
5 coords2 = sunpy.map.all_coordinates_from_map(sunspot2)
6 vels_m2 = deproj_vel(coords2.Tx[:196, :196], coords2.Ty[:196, :196], flows2.vx, flows2.vy)
7
8 vx_deproj2 = vels_m2[1]
9 vy_deproj2 = vels_m2[2]
10 vv = np.sqrt(vx_deproj2**2 + vy_deproj2**2)
11
```

EXIT> The spot image with LCT velocity field.

Conclutions

TODAY WE LEARNED

1. HOW TO WORK WITH CONTNUUM AND MAGNETOGRAM DATA
2. WORKING WITH A DATACUBE
3. TRACKING HORIZONTAL PLASMA FLOWS
4. DEFINITION OF MOAT FLOWS



NEXT SESSION › Dopplergrams and Helioseismology



Thanks!



Any question?

Juan Pablo Herrera Moreno
juapherreraamor@unal.edu.co



Registro
tiny.cc/escuelasolar

Más información
astronomiaoan.co/gosa/escuela.html



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